Dylan Spektor

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EDUCATION & HONORS

University of Colorado Boulder **Expected: April 2028**

B.S. Applied Mathematics & B.A. Computer Science

GPA: 4.0

Organizations: CU Quants, Engineering Honors Program, Tau Beta Pi, Quantum Scholars.

Relevant Coursework: Advanced Data Science, Neural Networks & Deep Learning, Software Development, Algorithms, Probability & Statistics, Applied Real Analysis, Numerical Methods.

Spring 2027 (Planned): Decision Making Under Uncertainty (grad-level), Markov Processes, Algorithmic Foundations of AI.

Colorado Early Colleges Douglas County North **May 2025**

A.S. Computer Science (Dual Enrollment at Arapahoe Community College) | 1580 SAT, National Merit Finalist

RESEARCH & QUANTITATIVE EXPERIENCE

Undergraduate Researcher, Programming Languages & Verification Group **Aug 2026 – Present**

Advised by Professor Bor-Yuh Evan Chang, CU Boulder

- Selected for a yearlong research apprenticeship in automated reasoning of software systems; targeting second authorship on a planned research submission.
- Implementing abstract interpretation engines in Lean 4, building and creating proofs for control-flow graph semantics to support a paper on demanded abstract interpretation of higher-order languages for fine-grained incremental analysis.

CU Quants Fund **Oct 2025 – Present**

Quantitative Developer

- Developing metric forecasting methods and historical order-book data pipelines to study stablecoin spreads, cryptocurrency market making, and perpetual-futures hedging.
- Tested crypto lead-lag signals using cross-correlation and transaction-cost assumptions; prototyped convex capital allocation across pairs strategies.

SELECTED PROJECTS & RESEARCH

Uncertainty-Aware Reinforcement Learning for LTL Tasks **Current**

- Proposed a DQN-based approach to LTL-guided reinforcement learning under detector miscalibration; will implement gridworld experiments benchmarking it against naive and uncertainty-aware baselines on task success and verification cost.

Spectral Bias in Neural PDE Solvers **Current**

- Investigating neural-network spectral bias in heat and wave equations via Fourier/Sturm–Liouville analysis and Neural Tangent Kernel (NTK) theory; studying the PDE-residual loss function of physics-informed neural networks (PINNs).

Exploratory Statistical Learning for Equity Pairs (PDF) **April 2026**

- Evaluated PCA pair screening and Engle-Granger cointegration across 497 S&P 500 equities; prototyped GMM regime clustering and MinHash-based historical pattern matching.

Virtrade | Full-Stack Paper Trading Application (GitHub) **April 2026**

- Collaborated in an agile team to build a full-stack paper trading application using Node.js/Express, Python/Flask microservices, and PostgreSQL. Integrated real-time market data APIs, interactive candlestick charts, a Bootstrap 5 UI, and session-based authentication.

SPX Options Arbitrage & Portfolio Liquidation (GitHub) **Oct 2026**

- Built a Python CLI to parse Cboe SPX/SPXW options chains and solve linear programs for static-arbitrage detection and portfolio liquidation; exports optimized results to CSV.

External Quantum Efficiency Degradation in Perovskite Cells **Dec 2025**

- In collaboration with the National Laboratory of the Rockies, analyzed 350–400 nm perovskite-cell EQE measurements across 0, 5, and 10 deposition cycles using bootstrap resampling, linear/quadratic regression, and ANOVA; found statistically significant differences in degradation coefficients.

TECHNICAL SKILLS & MISCELLANEOUS INTERESTS

Programming & Data: Python (NumPy, pandas, SciPy, scikit-learn, statsmodels, QuantLib), functional programming, Lean 4

Statistics & ML: probability, statistics, machine learning, deep learning, predictive modeling, time-series analysis, anomaly detection, regression, bootstrapping, hypothesis testing, cross-correlation, clustering, dimensionality reduction

Interests: Philosophy of language, AI safety, psychoanalysis, German idealism, mixed martial arts